

REVIEW OF RELATED LITERATURE: GAP MATRIX

Project FALCON: An Affordable Solar-Powered Smart Coastal Monitoring Buoy with Local Edge Processing and AI-Assisted Short-Term Wave-Height Prediction

Project FALCON Research Group · Fullbright College · August 2026

Purpose. This matrix compares related research, identifies what remains unresolved, and shows the specific contribution that Project FALCON can test. The gap column does not mean that prior studies are defective; it identifies differences in scope, environment, hardware, validation, or intended users.

Table 1. Related Literature, Research Gap, and Reference

TITLE	GAP	REF
A Low-Cost Sensor Buoy System for Monitoring Shallow Marine Environments Demonstrated an affordable autonomous wireless platform for shallow-water monitoring.	The system focused on environmental monitoring and communications, not locally executed short-horizon wave-height prediction, transparent model status, or GPS/IMU displacement assessment. FALCON can extend the integrated low-cost approach by evaluating these functions on an ESP32–Orange Pi architecture.	Albaladejo et al. (2012)
Observing Sea States Synthesized modern sea-state observation methods and the limitations of platforms and sensors.	The review establishes that a buoy's motion, response, sampling, and processing affect wave estimates, but it does not provide a student-scale implementation for a specific Philippine coastal site. FALCON must translate these principles into a calibrated, documented prototype and avoid treating raw IMU motion as wave height.	Ardhuin et al. (2019)
Measuring Waves with Pressure Transducers Explained how subsurface pressure may be converted to wave information and why correction is required.	Pressure attenuation, sensor depth, and high-frequency loss prevent pressure from being used as direct surface elevation. FALCON can contribute a pressure-assisted method only after depth-aware calibration and comparison against a reference measurement.	Bishop & Donelan (1987)
A Nonlinear Weakly Dispersive Method for Recovering the Elevation of Irrotational Surface Waves from Pressure Measurements Improved pressure-based free-surface reconstruction for nonlinear waves.	The method is mathematically rigorous but was not designed as a low-compute, real-time workflow for an inexpensive edge buoy. FALCON can evaluate whether a simplified, documented reconstruction is accurate and computationally practical on its selected hardware.	Bonneton et al. (2018)
A Novel Model to Predict Significant Wave Height Based on Long Short-Term Memory Network Showed that LSTM models can learn temporal patterns for significant-wave-	Results from established datasets do not prove performance using a new low-cost local buoy with limited, noisy records. FALCON must use chronological validation, compare against persistence and simple baselines, and report error, latency, and uncertainty for its 5–15 minute target.	Fan et al. (2020)

TITLE	GAP	REF
height prediction.		
Seawater Battery-Based Wireless Marine Buoy System with Battery Degradation Prediction and Multiple Power Optimization Capabilities Integrated energy awareness, communications, and load optimization for marine autonomy.	The study used a different power architecture and did not evaluate FALCON's proposed solar–LiFePO4 system or dual-computer load. FALCON must produce a measured energy budget, brownout behavior, charging profile, and reduced-sun endurance test.	Cho et al. (2021)
Beach Deployment of a Low-Cost GNSS Buoy for Determining Sea-Level and Wave Characteristics Demonstrated that low-cost GNSS can capture useful coastal sea-level and wave information.	The deployment was short and used post-processed GNSS in a different coastline. FALCON combines GNSS primarily for position and displacement monitoring with IMU/pressure wave sensing; its local accuracy and false-alarm behavior still require testing.	Knight et al. (2021)
Measurement Principle and Technology of Miniaturized Strapdown Inertial Wave Sensor Presented orientation correction, acceleration transformation, integration, and spectral processing for a compact inertial wave sensor.	The work validates the measurement principle, not the BNO085 or FALCON hull. FALCON must quantify orientation error, gravity removal, integration drift, sampling stability, hull response, and mooring influence through controlled and field comparison tests.	Zhou et al. (2022)
Prediction of Significant Wave Height Based on EEMD and Deep Learning Used signal decomposition and deep learning for nonlinear, non-stationary wave prediction.	The approach may be too computationally complex and data-hungry for a small edge computer and an early local dataset. FALCON can test whether a compact model offers a meaningful improvement over simpler baselines within power, memory, and latency limits.	Song et al. (2023)
Challenges, Limitations, and Measurement Strategies to Ensure Data Quality in Deep-Sea Sensors Identified drift, corrosion, power limits, communication problems, and missing references as persistent data-quality risks.	The paper provides broad quality strategies rather than a complete low-cost coastal implementation. FALCON can operationalize them through per-record quality flags, missing/stuck-value checks, cross-sensor plausibility, local buffering, maintenance logs, and prediction suppression when inputs are invalid.	Skålvik et al. (2023)
Monitoring Tides, Currents, and Waves along Coastal Habitats Using the Mini Buoy Validated reinforced low-cost accelerometer-based devices for inundation, currents, and wave orbital velocity.	The bottom-mounted Mini Buoy has a different mechanics and purpose from a surface-following, solar-powered telemetry buoy. FALCON must validate its own float response and long-term enclosure, mooring, UV, corrosion, and biofouling performance.	Ladd et al. (2024)

A Low-Cost Autonomous Multi-Functional Buoy for Ocean Currents and Seawater Parameter Monitoring, and Particle Tracking Reported a modular solar-powered buoy with local logging, satellite telemetry, GPS, and successful short field deployments.	The platform emphasizes drifting and water-quality observations, uses satellite communication, and does not test edge-based 5–15 minute wave prediction or multi-condition anti-displacement logic. FALCON targets a moored, locally maintainable and local-first coastal implementation.	Williams et al. (2025)
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Brief Review of Each Related Study

1. Albaladejo et al. (2012) — A Low-Cost Sensor Buoy System for Monitoring Shallow Marine Environments. The researchers developed an affordable wireless buoy for continuous monitoring in shallow marine environments. Their design combined sensing, embedded electronics, wireless communication, and autonomous power in one compact platform. The study showed that low-cost components can collect useful observations when stability, communication, and energy are planned as one system. This work supports FALCON's low-cost approach and reminds the researchers that the physical buoy, electronics, and software must operate reliably together.

2. Ardhuin et al. (2019) — Observing Sea States. This review examined the principal methods used to observe waves and sea conditions. It explained that satellites, radar, ships, and buoys provide different but complementary measurements. The authors emphasized that buoy hull response, mooring behavior, sampling, sensor noise, and processing can affect the reported wave values. This is important to FALCON because raw IMU movement cannot automatically be described as accurate wave height without calibration and reference comparison.

3. Bishop and Donelan (1987) — Measuring Waves with Pressure Transducers. The study investigated how underwater pressure records can be converted into information about surface waves. It showed that wave pressure weakens as sensor depth and wave frequency increase. The researchers explained that correction methods are required and that high-frequency wave information may be lost or distorted. This gives FALCON a scientific basis for pressure-assisted measurement, but the Bar02 data must be depth-corrected and compared with a reference instrument.

4. Bonneton et al. (2018) — A Nonlinear Weakly Dispersive Method for Recovering Surface-Wave Elevation from Pressure. The authors proposed a nonlinear method for reconstructing the water surface from pressure measured below it. Their approach addressed conditions in which a simple linear correction may not adequately describe the waves. The research demonstrated that water depth, dispersion, and wave nonlinearity affect reconstruction quality. FALCON can use these ideas when selecting its pressure-processing method, while testing whether the method is accurate and practical on a small edge computer.

5. Fan et al. (2020) — A Novel Model to Predict Significant Wave Height Based on Long Short-Term Memory Network. This work applied an LSTM neural network to significant-wave-height forecasting. LSTM models learn relationships across sequences of earlier observations and are commonly used for time-series prediction. The researchers demonstrated that historical wave patterns can support useful future estimates when adequate training data are available. The study supports FALCON's AI direction, but FALCON must still compare its model with persistence and simpler baselines using unseen chronological local data.

6. Cho et al. (2021) — Seawater Battery-Based Wireless Marine Buoy System with Power Optimization. The researchers designed a marine buoy whose communication and sensing activities were managed according to available energy and battery condition. They considered battery degradation, operating loads, and several power-saving strategies instead of assuming an unlimited energy supply. Their results illustrate why autonomous marine systems require active energy management. This is relevant to FALCON's solar-LiFePO₄ design because the ESP32, Orange Pi, sensors, and radio must be tested using a measured daily energy budget.

7. Knight et al. (2021) — Beach Deployment of a Low-Cost GNSS Buoy for Determining Sea-Level and Wave Characteristics. The researchers deployed a low-cost GNSS buoy in a coastal beach environment. They compared its

observations with nearby established instruments to determine whether inexpensive positioning hardware could provide useful sea-level and wave information. The results demonstrated the potential of GNSS while also showing the importance of suitable processing and reference validation. For FALCON, GNSS can support position and displacement monitoring, but normal receiver error and anchor swing must not be mistaken for theft or anchor failure.

8. Zhou et al. (2022) — Measurement Principle and Technology of a Miniaturized Strapdown Inertial Wave Sensor. This study described a compact inertial system that estimates waves from buoy motion. Its method used attitude information to transform acceleration into the correct reference frame before integration and spectral processing. The work explained that gravity removal, orientation correction, drift control, and stable sampling are necessary for meaningful wave parameters. It supports FALCON's use of an IMU, but the specific BNO085, enclosure position, hull, and mooring still require controlled validation.

9. Song et al. (2023) — Prediction of Significant Wave Height Based on EEMD and Deep Learning. The researchers combined ensemble empirical mode decomposition with deep learning for significant-wave-height prediction. Decomposition separated a complex and non-stationary wave record into simpler components that a model could learn. Their work showed the potential benefit of combining signal processing with neural networks for difficult ocean time series. FALCON may study a similar direction, but it must determine whether added accuracy justifies the computation, memory, data, and energy required on the Orange Pi.

10. Skålvik et al. (2023) — Challenges, Limitations, and Measurement Strategies to Ensure Data Quality in Deep-Sea Sensors. This paper examined common causes of unreliable long-term marine observations. It discussed calibration drift, corrosion, biofouling, limited power, unstable communication, and difficulty obtaining reference measurements. The authors emphasized that quality assurance must begin before deployment and continue throughout operation. FALCON should therefore attach health and quality flags to its records and prevent the AI feature from presenting invalid sensor input as a reliable prediction.

11. Ladd et al. (2024) — Monitoring Tides, Currents, and Waves along Coastal Habitats Using the Mini Buoy. The researchers improved and tested inexpensive accelerometer-based devices for coastal hydrodynamic monitoring. They calibrated the devices and compared their measurements with industry-standard instruments. The updated design improved physical strength and enabled observations of inundation, current velocity, and wave orbital motion. This work demonstrates both the value of affordable motion sensing and the need for FALCON to test mechanical durability, calibration, mooring effects, and environmental exposure.

12. Williams et al. (2025) — A Low-Cost Autonomous Multi-Functional Buoy for Ocean Currents, Seawater Monitoring, and Particle Tracking. This study developed a compact solar-powered buoy using modular sensors, local logging, GPS, and satellite telemetry. The system collected water-quality and position data during several short coastal deployments. The researchers reported strong data recovery and communication reliability, demonstrating the feasibility of affordable autonomous observation. Its focus differs from FALCON, which proposes a moored, local-first platform centered on wave monitoring, edge prediction, system health, and abnormal-displacement detection.

Interpretation: Existing literature supports the feasibility of each major subsystem, but no reviewed study validates the same complete combination, cost constraints, local-first architecture, and intended Palawan coastal use case proposed for FALCON.

Consolidated Research Gap

Prior studies establish that low-cost buoys, inertial or pressure-assisted wave measurement, GNSS monitoring, renewable power, and machine-learning wave prediction are individually feasible. However, the reviewed literature does not establish that these capabilities will remain accurate and dependable when integrated into a student-built, solar-powered buoy using inexpensive sensors, intermittent connectivity, and a small edge computer in a Philippine near-shore environment. In particular, there is limited evidence on the combined effects of hull and mooring response, sensor quality, local data scarcity, power constraints, model uncertainty, offline operation, and false displacement alarms within one reproducible low-cost platform.

Proposed Novel Contribution of Project FALCON

1. **Locally adapted integration:** A reproducible ESP32–Orange Pi coastal buoy designed around affordable, obtainable components and a defined Palawan test site.
2. **Calibrated wave observation:** Wave estimates derived from synchronized motion and pressure measurements, with hull/mooring effects and reference comparison explicitly documented.
3. **Honest edge prediction:** A clearly defined 5–15 minute wave-height target evaluated chronologically against persistence and other simple baselines, including error, latency, and uncertainty.
4. **Local-first resilience:** Onboard acquisition, quality checks, storage, and dashboard service that continue during internet outages and recover delayed records without silently duplicating data.
5. **Integrated health and displacement logic:** GPS, IMU, time persistence, sensor health, and power state combined to distinguish likely abnormal movement from ordinary anchor swing and GPS noise.
6. **Whole-system validation:** Measurement accuracy, energy autonomy, communication reliability, maintainability, and alert performance assessed together rather than presenting component integration alone as novelty.

Defensible novelty statement: Project FALCON is novel not because it invents the smart buoy, wave sensor, or machine-learning forecast, but because it designs and experimentally evaluates their affordable, local-first integration for a defined Philippine coastal setting, with transparent data quality, uncertainty, energy performance, and displacement-alert limitations.

APA 7th Edition References

- Albaladejo, C., Soto, F., Torres, R., Sánchez, P., & López, J. A. (2012). A low-cost sensor buoy system for monitoring shallow marine environments. *Sensors*, 12(7), 9613–9634. <https://doi.org/10.3390/s120709613>
- Ardhuin, F., Stopa, J. E., Chapron, B., Collard, F., Husson, R., Jensen, R. E., Johannessen, J., Mouche, A., Passaro, M., Quartly, G. D., Swail, V., & Young, I. R. (2019). Observing sea states. *Frontiers in Marine Science*, 6, Article 124. <https://doi.org/10.3389/fmars.2019.00124>
- Bishop, C. T., & Donelan, M. A. (1987). Measuring waves with pressure transducers. *Coastal Engineering*, 11(4), 309–328. [https://doi.org/10.1016/0378-3839\(87\)90031-7](https://doi.org/10.1016/0378-3839(87)90031-7)
- Bonneton, P., Lannes, D., Martins, K., & Michallet, H. (2018). A nonlinear weakly dispersive method for recovering the elevation of irrotational surface waves from pressure measurements. *Coastal Engineering*, 138, 1–8. <https://doi.org/10.1016/j.coastaleng.2018.04.005>
- Cho, J., Kim, M. W., Kim, Y., Park, J.-S., Lee, D.-H., Kim, Y., & Kim, J. J. (2021). Seawater battery-based wireless marine buoy system with battery degradation prediction and multiple power optimization capabilities. *IEEE Access*, 9, 104104–104114. <https://doi.org/10.1109/ACCESS.2021.3098846>
- Fan, S., Xiao, N., & Dong, S. (2020). A novel model to predict significant wave height based on long short-term memory network. *Ocean Engineering*, 205, Article 107298. <https://doi.org/10.1016/j.oceaneng.2020.107298>
- Knight, P. J., Bird, C. O., Sinclair, A., Higham, J., & Plater, A. J. (2021). Beach deployment of a low-cost GNSS buoy for determining sea-level and wave characteristics. *Geosciences*, 11(12), Article 494. <https://doi.org/10.3390/geosciences11120494>
- Ladd, C. J. T., Burden, A., Skov, M. W., & van der Wal, D. (2024). Monitoring tides, currents, and waves along coastal habitats using the Mini Buoy. *Limnology and Oceanography: Methods*, 22(4), 259–279. <https://doi.org/10.1002/lom3.10631>
- Skålvik, A. M., Saetre, C., Frøysa, K.-E., Bjørk, R. N., & Tengberg, A. (2023). Challenges, limitations, and measurement strategies to ensure data quality in deep-sea sensors. *Frontiers in Marine Science*, 10, Article 1152236. <https://doi.org/10.3389/fmars.2023.1152236>
- Song, T., Wang, J., Huo, J., Wei, W., Han, R., Xu, D., & Meng, F. (2023). Prediction of significant wave height based on

EEMD and deep learning. *Frontiers in Marine Science*, 10, Article 1089357. <https://doi.org/10.3389/fmars.2023.1089357>

Williams, Z., Soto Calvo, M. A., Lee, H. S., Aljber, M., & Jeong, J.-S. (2025). A low-cost autonomous multi-functional buoy for ocean currents and seawater parameter monitoring, and particle tracking. *Journal of Marine Science and Engineering*, 13(9), Article 1629. <https://doi.org/10.3390/jmse13091629>

Zhou, F., Zhang, R., & Zhang, S. (2022). Measurement principle and technology of miniaturized strapdown inertial wave sensor. *Frontiers in Marine Science*, 9, Article 991996. <https://doi.org/10.3389/fmars.2022.991996>

Prepared for academic discussion. Final claims of accuracy, prediction capability, autonomy, and field readiness remain subject to calibration and physical field validation.